Lecture notes: Sorting algorithms (SAMPLE)

Bubble sort repeatedly swaps adjacent elements that are out of order. It runs in O(n^2) time.
Insertion sort builds a sorted prefix by inserting each element into its correct position.
It is O(n^2) in the worst case but O(n) on an already sorted array.
Merge sort splits the array in half, sorts each half recursively, and merges the halves.
Merge sort runs in O(n log n) time and uses O(n) extra space; it is stable.
Quicksort chooses a pivot, partitions the array around it, and sorts each side recursively.
Quicksort is O(n log n) on average but O(n^2) in the worst case, for example when the pivot
is always the smallest element. Choosing a random pivot makes the worst case unlikely.
A sorting algorithm is stable if equal elements keep their original relative order.
Any comparison-based sorting algorithm needs at least n log n comparisons in the worst case.

Binary search finds an element in a sorted array in O(log n) time by halving the range.